

Industrial Internship Report on "Password Manager using Python"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "Password Manager using Python." The objective of this project was to develop a secure desktop application that allows users to generate, store, search, view, and delete passwords efficiently. The application was developed using Python with Tkinter for the graphical user interface, SQLite for database management, and the Cryptography (Fernet) library for encrypting passwords before storing them. This project helped me gain practical experience in GUI development, database integration, encryption techniques, software testing, and secure application development while addressing the real-world challenge of managing passwords safely.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

The Industrial Internship offered by upskill Campus in collaboration with The IoT Academy and UniConverge Technologies Pvt. Ltd. (UCT) provided an excellent opportunity to gain practical experience in Python application development. The internship focused on developing real-world problem-solving skills by designing and implementing a complete software project within the given timeline. This program helped bridge the gap between academic learning and industry expectations by providing hands-on exposure to software development practices.

As part of this internship, I successfully developed a Password Manager using Python. The objective of the project was to create a secure desktop application that enables users to generate, store, search, view, and delete passwords efficiently. The application was developed using Python, Tkinter for the graphical user interface, SQLite for database management, and the Cryptography (Fernet) library for secure password encryption. This project enhanced my understanding of GUI development, database integration, encryption techniques, and application testing.

The internship was well structured and divided into weekly milestones. During the initial weeks, I understood the project requirements and designed the application. In the following weeks, I implemented the core functionalities, integrated password encryption, performed testing, improved the user interface, and completed project documentation. Regular weekly progress reports helped me monitor my work and maintain consistent progress throughout the internship.

This internship significantly improved my technical knowledge as well as my problem-solving, debugging, and project management skills. It also increased my confidence in developing complete software applications independently while following an organized development process.

I sincerely express my gratitude to upskill Campus, The IoT Academy, and UniConverge Technologies Pvt. Ltd. (UCT) for providing this valuable internship opportunity. I also thank the internship mentors, my faculty members, and everyone who supported and guided me throughout this learning journey.

Finally, I would encourage my juniors and fellow students to actively participate in industrial internships. Such opportunities provide valuable practical exposure, improve technical skills, strengthen project development experience, and prepare students for future careers in the software industry.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



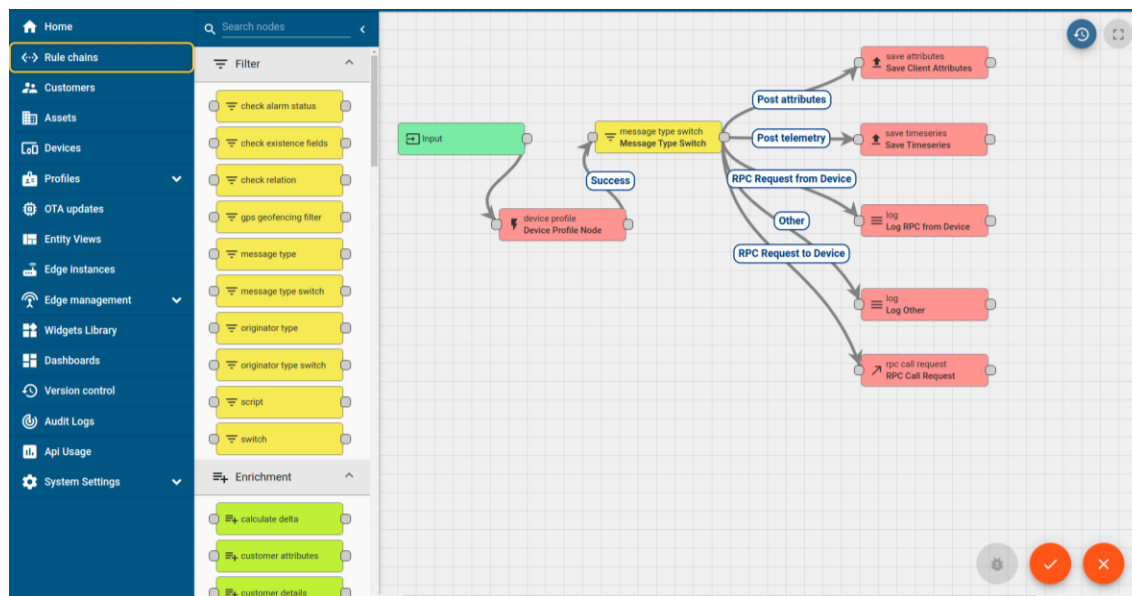
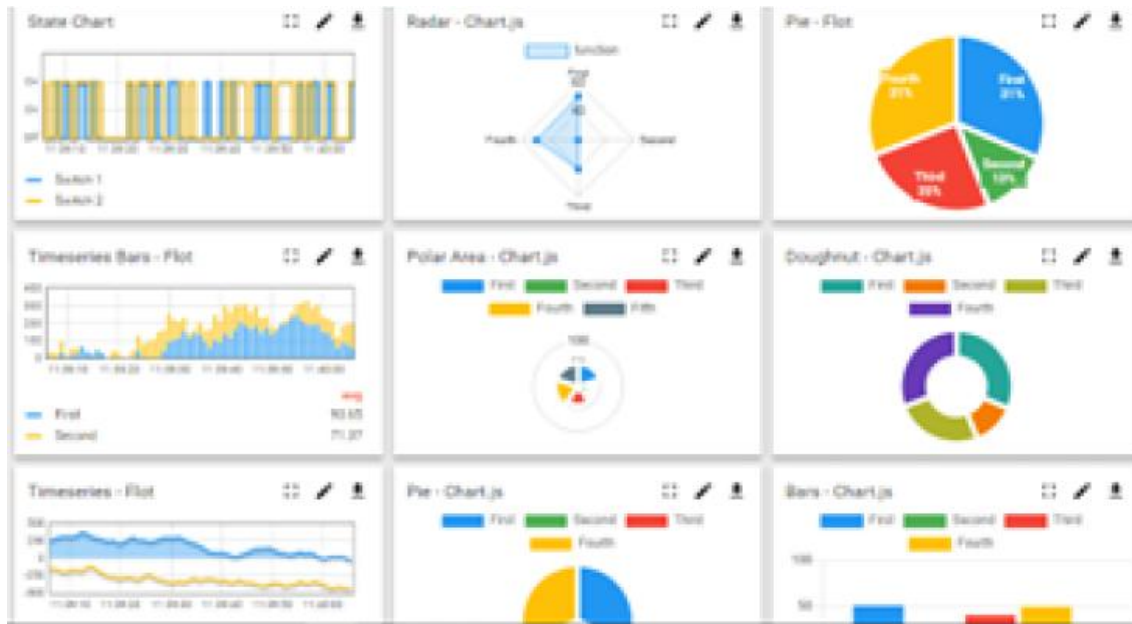
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleashed the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



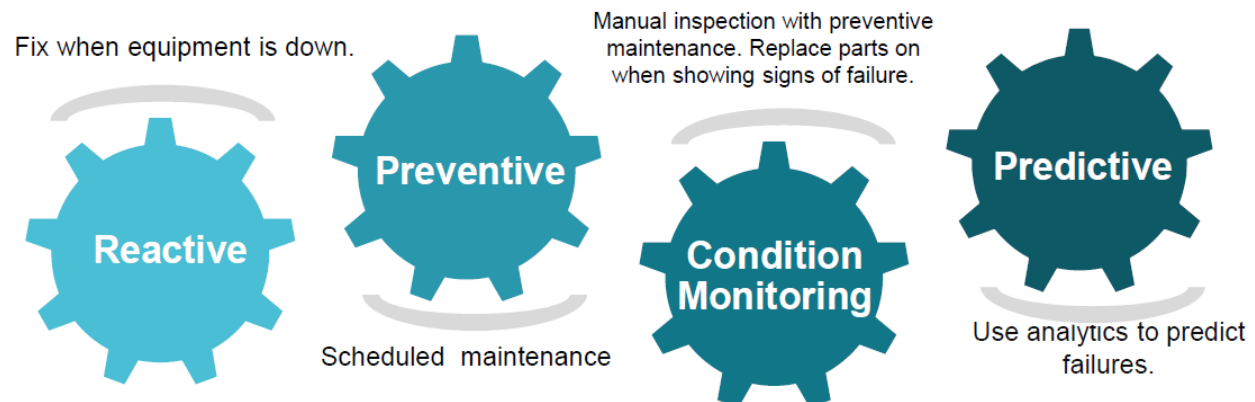


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

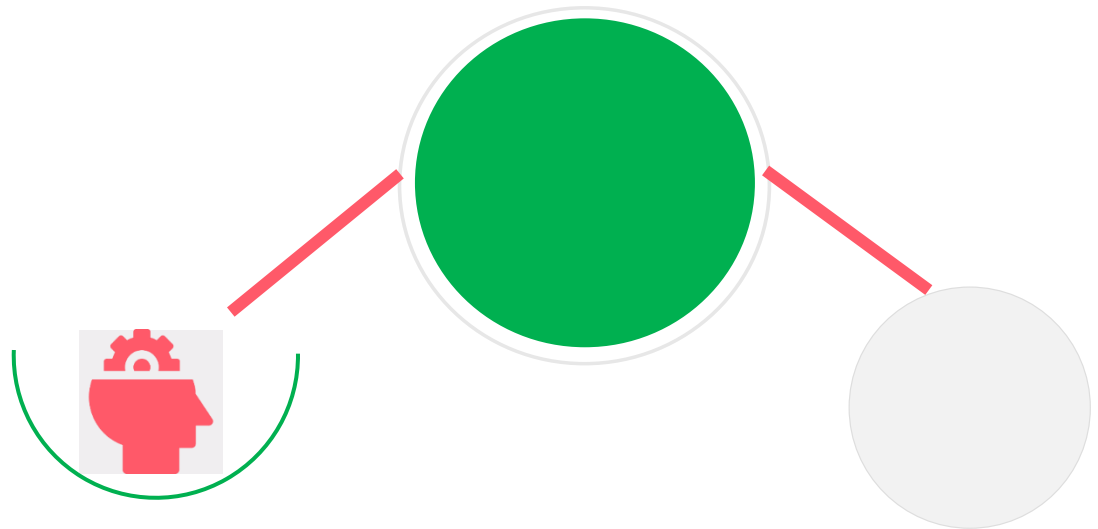
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

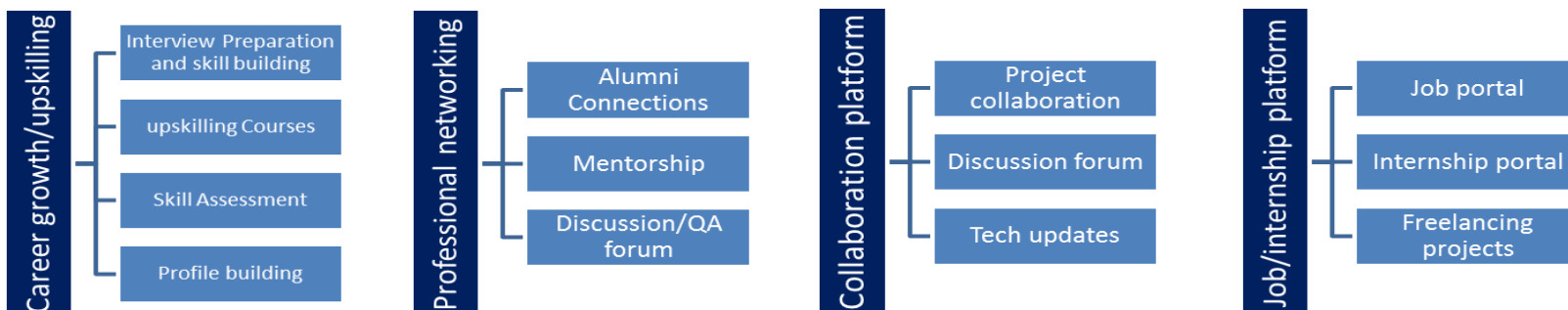
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- To develop a user-friendly desktop application using the Tkinter GUI framework.
- To generate strong and random passwords for improved account security.
- To securely store website credentials using SQLite as the database.
- To protect stored passwords using Fernet encryption from the Python Cryptography library.
- To provide functionalities such as Save Password, View Passwords, Search Password, and Delete Password.
- To implement proper input validation and error handling for reliable application performance.
- To gain practical experience in Python application development, database management, encryption techniques, GUI design, and software testing.
- To understand the complete software development lifecycle, including requirement analysis, implementation, testing, debugging, documentation, and project deployment.

2.5 Reference

[1] Python Software Foundation. Python 3 Documentation.

<https://docs.python.org/3/>

[2] Python Software Foundation. tkinter — Python interface to Tcl/Tk.

<https://docs.python.org/3/library/tkinter.html>

[3] SQLite Documentation. SQLite Documentation.

<https://www.sqlite.org/docs.html>

[4] Cryptography Project. Fernet (Symmetric Encryption).

<https://cryptography.io/en/latest/fernet/>

[5] W3Schools. Python Tutorial.

<https://www.w3schools.com/python/>

[6] GeeksforGeeks. Python Programming Language.

<https://www.geeksforgeeks.org/python-programming-language/>

2.6 Glossary

Terms	Acronym
Python	A high-level, object-oriented programming language used to develop the Password Manager application.
Tkinter	Python's standard GUI library used to create the graphical user interface of the application.
SQLite	A lightweight relational database used to store website credentials securely.
Cryptography	A Python library used to encrypt and decrypt stored passwords for enhanced security.
GUI (Graphical User Interface)	The visual interface through which users interact with the Password Manager application.
Database	A structured storage system used to save website names, usernames, and encrypted passwords.
Encryption	The process of converting readable information into a secure format to protect sensitive data.
Decryption	The process of converting encrypted data back into its original readable form.
Fernet	A symmetric encryption method provided by the Cryptography library to securely encrypt and decrypt passwords.

3 Problem Statement

The assigned project for this internship was to develop a Password Manager using Python that securely stores and manages user credentials. The increasing number of online accounts makes it difficult for users to remember multiple strong passwords, leading many to reuse passwords or store them in insecure places such as text files or notebooks.

The objective of this project is to provide a secure and user-friendly desktop application that allows users to generate strong passwords, save login credentials, search for stored passwords, view all saved records, and delete passwords when they are no longer needed.

The application was developed using Python with Tkinter for the graphical user interface, SQLite for database management, and the Cryptography (Fernet) library to encrypt passwords before storing them. This ensures that sensitive user information remains protected while providing quick and convenient access to saved credentials.

The Password Manager offers a simple, efficient, and secure solution for managing passwords, helping users improve their online security while reducing the risk of password-related security issues.

4 Existing and Proposed solution

Existing Solution

Many users store passwords in notebooks, text files, spreadsheets, or reuse the same password across multiple accounts. Although web browsers provide password-saving features, these methods have security limitations and do not always offer complete control over stored credentials.

Limitations of Existing Solutions

- Passwords stored in text files are not encrypted.
- Reusing the same password across different websites reduces security.
- Manual password management is difficult and time-consuming.
- Some password storage methods lack proper search and management features.
- Users may forget complex passwords or store them in insecure locations.

Proposed Solution

The proposed solution is a Password Manager using Python that securely stores and manages user credentials. The application allows users to generate strong passwords, save encrypted credentials, search, view, and delete passwords through a simple graphical user interface. The application is developed using Python, Tkinter, SQLite, and the Cryptography (Fernet) library to ensure secure password storage and easy password management.

Value Addition

- Secure password storage using Fernet encryption.
- Automatic generation of strong and random passwords.
- Easy-to-use graphical user interface developed using Tkinter.
- Efficient storage and retrieval of credentials using SQLite.
- Quick search functionality to find saved passwords.
- Ability to view, save, and delete password records.

4.1 Code submission (Github link):

Project Repository: <https://github.com/Jhahnavi30/upskillcampus>

Project Code: <https://github.com/Jhahnavi30/upskillcampus/blob/main/main.py>

4.2 Report submission (Github link):

https://github.com/Jhahnavi30/upskillcampus/blob/main/PasswordManager_JhahnaviKP_USC_UC T.pdf

5 Proposed Design/ Model

The proposed Password Manager is designed as a secure desktop application developed using Python. The application follows a simple workflow that begins with user input through the graphical interface and ends with the secure storage or retrieval of encrypted passwords. The system consists of three main components: the graphical user interface (Tkinter), the encryption module (Fernet Cryptography), and the SQLite database. Initially, the user enters the website name, username, and password or generates a strong password using the password generator. The application validates the input, encrypts the password, and stores it securely in the database. Whenever the user searches or views stored credentials, the encrypted password is retrieved from the database, decrypted, and displayed through the interface. This design ensures secure password management while maintaining a simple, lightweight, and user-friendly application.

5.1 High Level Diagram (if applicable)

The high-level architecture of the Password Manager consists of four major components: User, Graphical User Interface (Tkinter), Application Logic, and SQLite Database. The user interacts with the application through the graphical interface to enter website credentials or generate secure passwords. The application validates the user input and encrypts passwords using the Fernet Cryptography library before storing them in the SQLite database. Whenever a user searches or views saved credentials, the encrypted password is retrieved from the database, decrypted, and displayed through the graphical interface. This architecture ensures secure storage, efficient data management, and an easy-to-use desktop application.

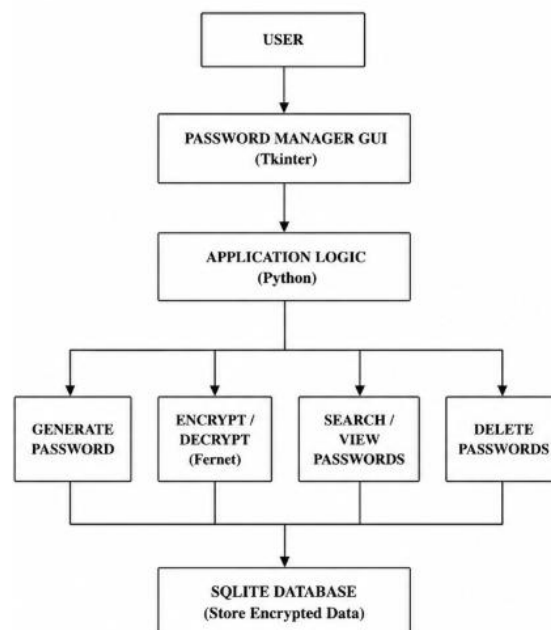


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

The low-level workflow describes the internal operation of the Password Manager application. Initially, the user enters the website name, username, and password through the graphical interface. The application validates the input data and generates a strong password if requested by the user. Before storing the password, the system encrypts it using the Fernet encryption algorithm. The encrypted credentials are then saved in the SQLite database. When the user performs a search or requests to view saved credentials, the encrypted password is retrieved from the database, decrypted using the same encryption key, and displayed through the graphical interface. Users can also delete stored records whenever required. This workflow ensures secure password storage and efficient management of user credentials.

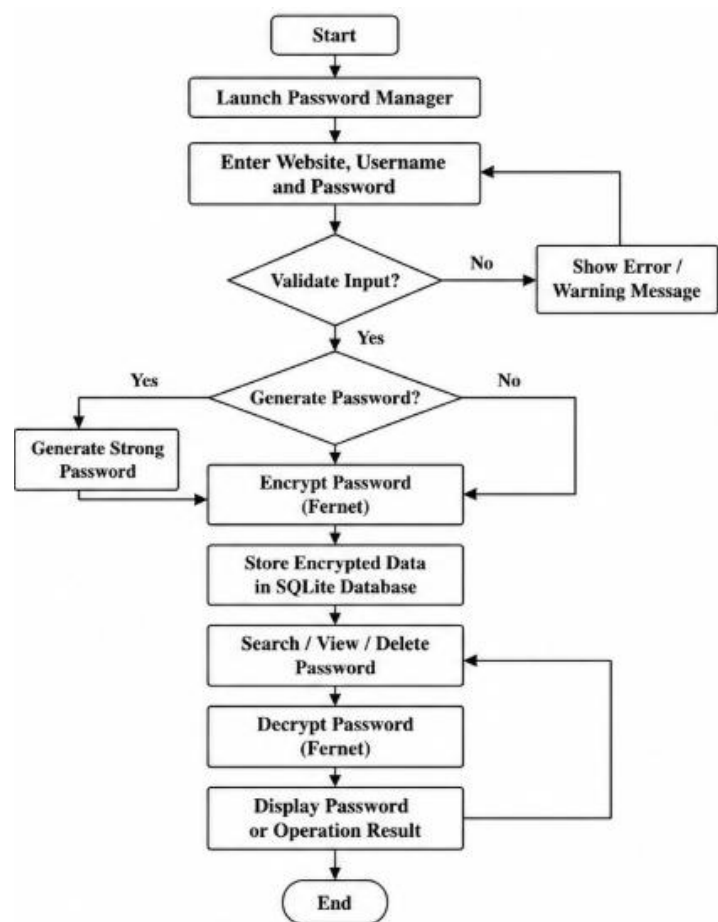


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces (if applicable)

The Password Manager application provides a simple and user-friendly graphical interface developed using the Tkinter library. The interface enables users to securely manage their passwords through different functional modules.

The application interface consists of the following components:

- Website Field – Allows the user to enter the website or application name
- Username Field – Accepts the username associated with the website
- Password Field – Allows users to enter or generate a secure password
- Generate Password Button – Creates a strong random password
- Save Password Button – Encrypts and stores the password in the SQLite database
- View Passwords Button – Displays all saved credentials after decrypting them
- Search Password Button – Retrieves credentials based on the entered website name
- Delete Password Button – Removes selected credentials from the database

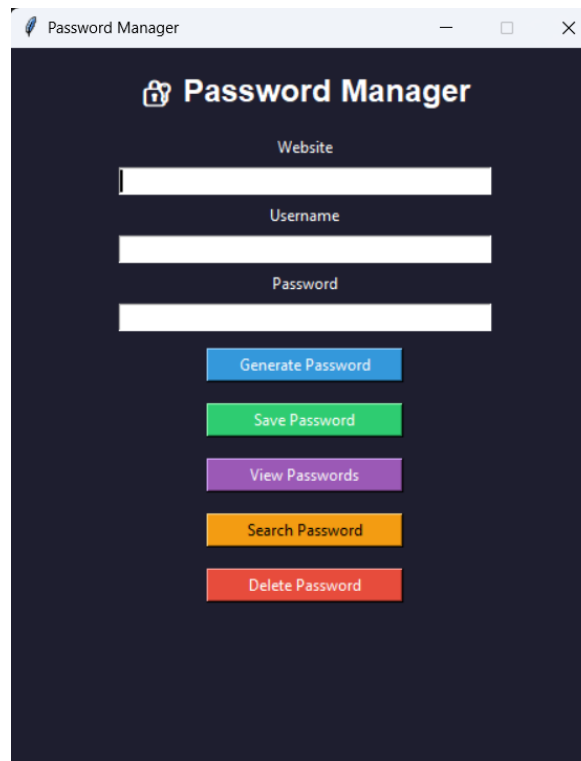


Figure 3: PASSWORD MANAGER MAIN INTERFACE

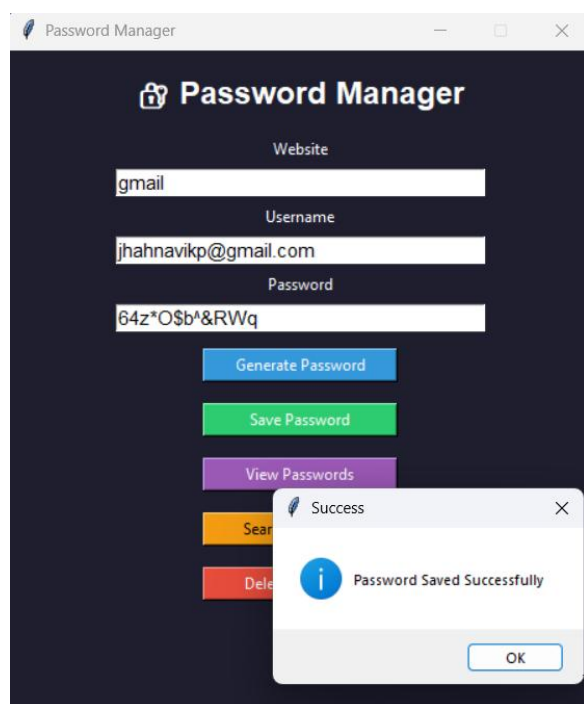


Figure 4: PASSWORD SAVED SUCCESSFULLY

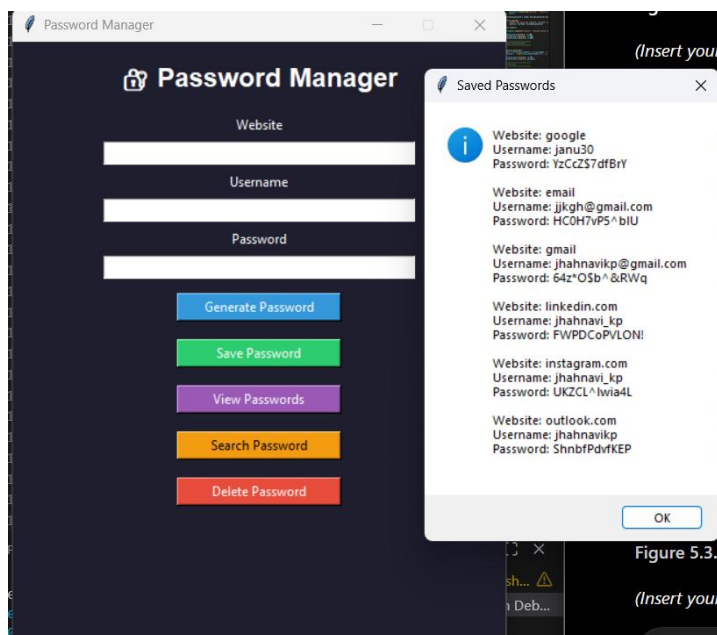


Figure 5: PASSWORD MANAGER MAIN INTERFACE

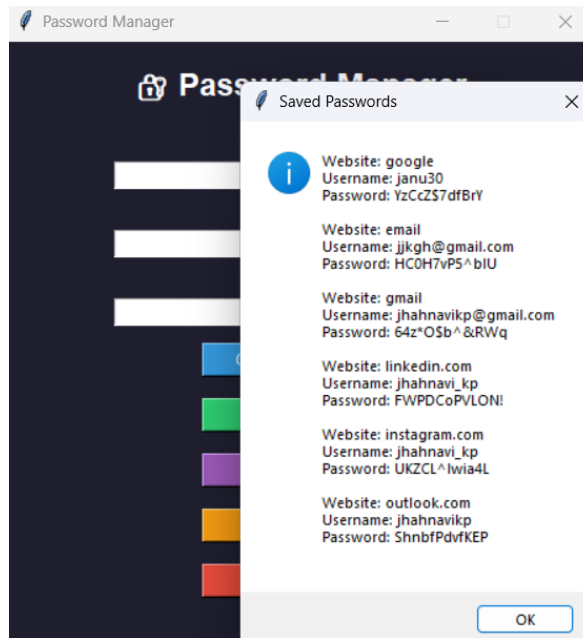


Figure 6: VIEWING STORED PASSWORDS

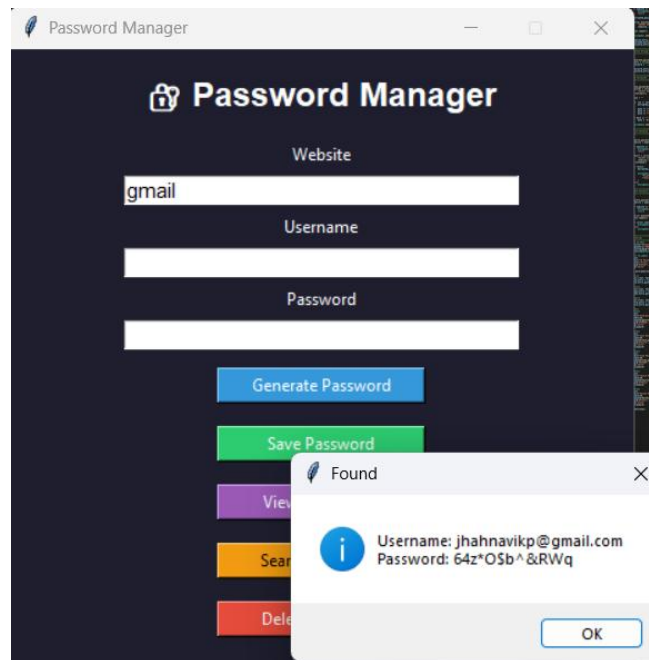


Figure 7: SEARCHING PASSWORD BY WEBSITE

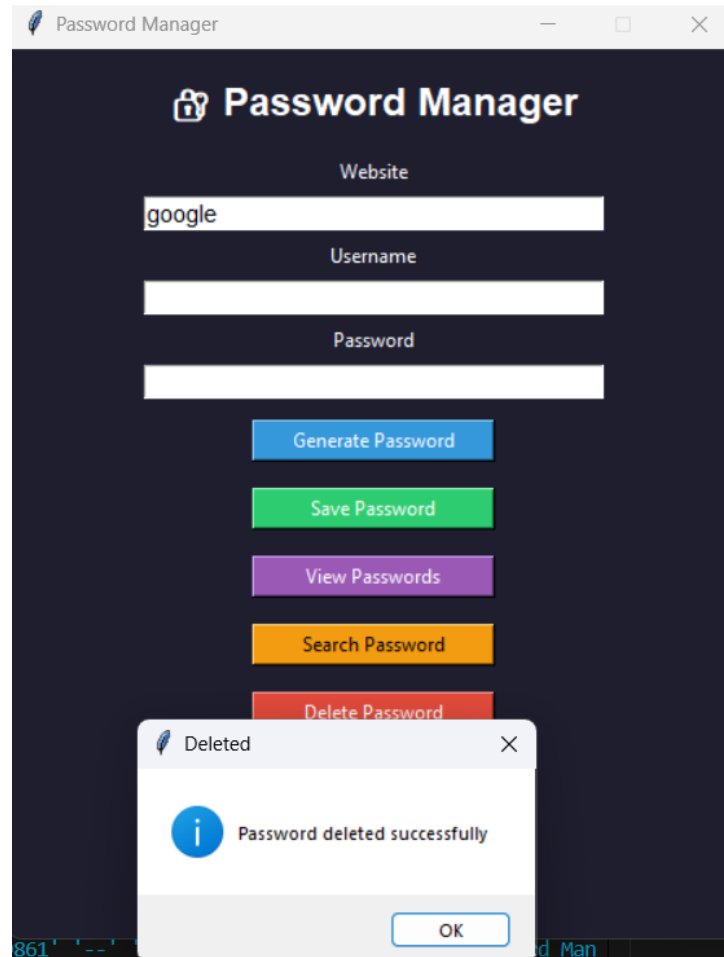


Figure 8: PASSWORD DELETED SUCCESSFULLY

6 Performance Test

The Password Manager application was tested to ensure reliable performance, secure password management, and smooth user interaction. Since the application is a desktop-based software system, the primary focus was on functionality, data security, response time, and database operations rather than hardware-related parameters such as power consumption or processor performance.

Several constraints were identified during the testing phase. The security of stored passwords was addressed by encrypting all passwords using the Fernet Cryptography library before storing them in the SQLite database. The application was also designed to validate user inputs, preventing empty fields from being stored in the database. Database operations such as saving, searching, viewing, and deleting records were tested repeatedly to ensure data consistency and accuracy.

The application demonstrated fast response times while performing password operations on a local SQLite database. Since the application is intended for individual desktop usage, memory consumption and processing requirements remained minimal. The testing confirmed that the application performs efficiently while maintaining secure password storage and a simple user experience.

Future improvements may include support for cloud synchronization, user authentication with a master password, automatic backup and recovery, and multi-device accessibility.

6.1 Test Plan/ Test Cases

Test Case	Expected Result	Actual Result	Status
Launch Application	Application opens successfully	Application launched correctly	Pass
Generate Password	Strong random password generated	Password generated successfully	Pass
Save Password	Credentials saved in database	Data stored successfully	Pass
View Passwords	Display all saved credentials	Records displayed correctly	Pass
Search Password	Retrieve password by website	Correct record displayed	Pass
Empty Input Validation	Display warning message	Warning displayed correctly	Pass
Password Encryption	Store encrypted password	Password stored securely	Pass

6.2 Test Procedure

The application was tested by executing each functionality individually and verifying the expected output. Sample website names, usernames, and passwords were entered through the graphical interface. Generated passwords were saved in the SQLite database after encryption using the Fernet algorithm. The stored credentials were retrieved using the Search and View functionalities to verify successful decryption. The Delete feature was tested by removing existing records from the database. Validation checks were also performed by attempting to save records with empty input fields to ensure that appropriate warning messages were displayed. All functionalities were tested multiple times to verify the application's reliability and correctness.

6.3 Performance Outcome

The testing results confirmed that the Password Manager application performs efficiently and reliably under normal usage conditions. All major functionalities, including password generation, encryption, database storage, retrieval, search, viewing, and deletion, worked as expected without runtime errors. The application successfully maintained data integrity while ensuring secure storage of user credentials. The graphical user interface remained responsive throughout testing, and database operations were completed quickly. Overall, the application met the intended project objectives and demonstrated the successful implementation of a secure desktop-based Password Management System.

7 My learnings

You The Industrial Internship provided me with valuable practical exposure to software development using Python. Throughout the internship, I gained hands-on experience in designing, developing, testing, and documenting a complete desktop application. Developing the Password Manager project helped me understand how theoretical programming concepts are applied to solve real-world problems.

During the internship, I improved my knowledge of Python programming, Tkinter GUI development, SQLite database management, and password encryption using the Cryptography (Fernet) library. I also learned how to implement secure data storage, validate user inputs, handle database operations, and debug application errors. Working on the project strengthened my understanding of the complete software development lifecycle, including requirement analysis, implementation, testing, documentation, and project deployment.

Preparing weekly progress reports and project documentation improved my technical writing, planning, and time management skills. I also gained confidence in independently developing a software application and presenting my work in a structured and professional manner.

Overall, this internship enhanced both my technical and professional skills. The knowledge and practical experience gained during this internship will help me in future academic projects, software development roles, and my overall career growth in the field of Computer Science and Software Engineering.

8 Future Scope

Although the Password Manager application successfully meets its primary objectives, there are several enhancements that can be implemented in future versions to improve its functionality, security, and user experience.

Some possible future improvements include:

- **Master Password Authentication:** Introduce a master password or biometric authentication to restrict unauthorized access to the application.
- **Password Update Feature:** Allow users to modify existing usernames and passwords without deleting and recreating records.
- **Password Visibility Toggle:** Provide an option to show or hide passwords while entering or viewing credentials.
- **Cloud Backup and Synchronization:** Enable secure cloud storage and synchronization of password data across multiple devices.
- **Import and Export Functionality:** Allow users to import existing passwords from files and export encrypted password data for backup purposes.
- **Password Strength Indicator:** Display password strength in real time and provide suggestions for creating stronger passwords.
- **Automatic Backup and Recovery:** Implement automatic database backup and recovery mechanisms to prevent accidental data loss.
- **Cross-Platform Compatibility:** Extend the application to support web and mobile platforms, allowing users to access their passwords from different devices.
- **Multi-User Support:** Enhance the application to support multiple user accounts with separate encrypted password databases.